Assignment Reflection

The project has given me insight into how several Python ideas could be integrated to build a full command-line application. Some of the concepts that I have employed while creating the Grade Tracker include variable, string, conditionals, loop, function, list, dictionary, class, inheritance, and exception handling. The most valuable thing that I have learned is the way to use object-oriented programming to structure the program. In the Grade Tracker program, there is an Assignment parent class and Homework and Exam child classes. In addition, I have used the GradeTracker class to organize the assignments as well as perform such operations as filtering and calculating the grade summary.

Among the problems that I faced while working on the project, is the use of indentation and class methods. Sometimes, the indentation was not right and that led to errors in Python, especially while implementing the summary features. It has been clear to me to read the Python error messages since they usually tell you the name of the file and the number of the line that had problems. Another problem that I had was related to using Git to commit my code since I committed from the wrong directory.

Input validation was another significant aspect of this assignment. It became evident that a computer application cannot take for granted the fact that the input given by users would be always correct. Input validation was applied to prevent input errors such as negative scores, scores higher than the highest possible score, empty subjects and titles, scores in non-numeric format, and incorrect date format.

Since i don’t have a lot of experience, improvements in the Grade Tracker would include permanent storage of data so that all of the tasks entered into the database would be still available once the application is closed.

Overall, this assignment helped to learn more about Python programming and the way individual programming concepts operate when an application is created.